

Tradelatest Book — RESEARCH (Grok)

Part VII extract: Program 1-9 results data table, research platform map, artifact roots under results/research/, Measurement Contract gap.

Generated 2026-08-07 from docs/book/
Grok review pass export — reading convenience only.
On conflict, repository code/config wins.

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Chapter 19 - The Research Programs: Falsification as a Discipline

Part VII - Research

Status of this chapter: Written (research data table expanded Grok pass)

Why this chapter exists

Chapter 1 promised that pattern interpreters and, by extension, trading ideas in general are "measured, never asserted." This chapter is where that promise gets tested at scale: a numbered sequence of Research Programs, each pre-registered, each run to a decisive conclusion, and - almost without exception so far - each one closing as an honest null result. That's not a failure of the research effort. It's the effort working exactly as designed.

What problem it solves

Gives a reader the shape of the research history **with concrete result data** (not only slogans): what was tested, sample sizes / key metrics where registered, the verdict, and why "0 PROMOTE" across program after program is a *credible* finding rather than a broken research pipeline.

What you need to already know

Chapter 16's Authority Ladder - every program below is that ladder applied to a specific hypothesis: information existing is not the same as economic value, which is not the same as earned authority.

The idea

Research results at a glance (data table)

Primary authority for every row: docs/current-findings.md (findings F-0xx below). Shared measurement standard for economic arms unless noted: **intraday fixed exit model + 12 bps** round-trip cost, absolute expectancy gate (E must clear 0 OOS to PROMOTE), plus controls / permutations as each program pre-registered. All economic cells below are **research-only** - none of these results grants production fusion weight.

Prog	Question (one line)	Key data (registered)	Verdict	Finding
1	Any next-bar directional edge on crypto majors?	Toys: baseline_delta vs random ~0 (-0.076...+0.033); spine 5-13 trades/inst -> n<30; pooled n=30 E=-0.399	0 PROMOTE / KILLED	F-019
1b	Conditional pockets (sessionxvolxmom, h<=20)?	25/25 partitions BH-significant but IG~0.001-0.0036 bits; Stage-3 economics on 74 cells -> 0 pockets	ENTROPY_LEAD_NO ECON	F-020

1c	Spine RETEST selection skill beyond session?	Selected–rejected $\Delta E = +1.17R$ but SESSION=ALL rejects; ZONE 0/110; SCORE $p = 0.327$	Session filter only	F-021
1d	Exit/cost grid rescue expectancy?	Universe $n = 53,629$; every SLxTP cell $E_{oos} < 0$; best -0.271 ; reality_gap $+4.16R$ (entry info, not exit)	Exit is not the lever	F-025
2	Completed sweep->disp->retest > bare sweep?	Funnel retest $n = 47$ (~1% completion); excursion_asym(h8) = -0.319 ; loses all 4 controls	INSUFFICIENT_POWER / FROZEN	F-026
3	H1/H4 rescue directional edge?	0 PROMOTE H1 and H4; H1 E_{oos} $-0.136 \dots -0.266$; H4 expansion_breakout ~gross 0 then pooled -0.0009	CLOSED null	F-027
P&F	Point-and-figure double-top/bottom edge?	BNB $n = 8554$, PF=0.528, $E = -0.454$, WR=0.327; loses random_uniform $\Delta E = -0.039$	REJECT / FROZEN	F-028
4	Vol-regime LEVEL consumable directionally?	$H_{atr} = 0.885$ (vol memory real); 0 REGIME_EXPLOITABLE ; toys $n = 3.7k-35k$ still $E \sim -0.14 \dots -0.35$	Informative != money	F-030
4b	Markov P^H transition forecast?	0 REGIME_EXPLOITABLE / 15 scopes; toys REGIME_REDUNDANT vs level+lag	CLOSED redundant	F-043
4d	MTF compression->expansion (Stage1 info / Stage2 econ)?	Stage1 UNIVERSAL $p \sim 0.0005$; Stage2 directional consumer 0 PROMOTE (crypto $E -0.25 \dots -0.60R$)	Info yes / money no	F-040
5	Cross-sectional relative-value monetizable?	5 interpreters, $n = 722-8,758$; momentum PF=0.335; 0 PROMOTE	KILLED (panel axis)	F-032
6	Carry/basis as signal on spot dispersion?	8 interpreters $n = 722-4,373$; $E \in [-0.0032, -0.0016]$; 0 PROMOTE	KILLED	F-033

6b	Funding cashflow harvest net of costs?	4 harvest_full REJECT n=104-729; pre-cost +1...+6 bps < ~24 bps RT cost	KILLED / STOP	F-034
FX	Program-1 null generalizes off crypto?	5 FX majors; toys n=7.7k-17k PF 0.02-0.10 E -1.5...-2.8R; 0 PROMOTE	Null generalizes	F-035
8	Weekly Mon-Fri liquidity-sweep ontology (FX)?	6 scopes; pooled n=512 E=-0.634R PF=0.457 ; 0 PROMOTE	CLOSED null	F-042
7	Open Interest	Named out-of-scope for P6; not run in findings ledger as of book sources	NOT RUN	-
9	M5 multi-TF expansion + OCO straddle	Pre-reg frozen; on-disk results/research/m5_mtf/ has resample_parity PASS only; Stage-1/2 verdicts not sealed into findings	PRE-REGISTERED	H-016 / prereg

How to read the table: "0 PROMOTE" means *no cell cleared the fixed absolute-expectancy bar under the pre-registered cost/exit model* - not "the research code failed." Several rows are statistically informative (entropy, vol memory, Stage-1 transition IG) yet economically dead for spot directional (or carry) constructions. That split is the point of the Authority Ladder.

Where the raw artifacts live (examples; full paths in each finding's Evidence field):

Family	Typical artifact roots
Program 1 qualify	results/research/qualification/
Conditional entropy	results/research/phase_b/
Selection effect	results/research/phase_s/
Exit grid	results/research/phase_d/
Structural asymmetry	results/research/phase_e/
HTF qualify	results/research/qualification_htf/
Regime / transition	results/research/regime/, results/research/candle_state/
Cross-section / carry	results/research/cross_sectional_*.json, carry_*.json, harvest_*.json
Weekly sweep	results/research/weekly_sweep/
Program 9 M5	results/research/m5_mtf/

The gate every program runs through

Nearly every program in this ledger is tested against the same standard: a pre-registered hypothesis, an intrabar-realistic exit model with real transaction costs (not close-only fills), and a decision rule that requires *absolute* expectancy above zero out-of-sample - not just beating a control. A program is only allowed to "PROMOTE" a finding into production authority if it clears that bar; anything else is REJECT, INSUFFICIENT_POWER, or REDUNDANT (statistically real but fully explained by something already known).

The ladder, program by program (narrative)

- **Program 1 - Next-Bar Directional Ontology.** The foundational sweep: is there *any* directional edge in next-bar entries across crypto majors, at any conditioning level (session, volatility, momentum)? Ran through qualification, conditional-entropy pocket search, a selection-effect isolation, a full trade-anatomy audit, and an exit/cost grid - every stage closed null. **CLOSED / KILLED.** This is the foundation the rest of the ladder builds on.
- **Program 2 - Structural Asymmetry.** Does a *completed* sweep->displacement->retest sequence add forward asymmetry beyond a bare sweep? One experiment, a negative point estimate, badly underpowered (n~47, roughly 1% funnel completion). **FROZEN after one experiment** - explicitly not run further until power improves.
- **Program 3 - Higher-Timeframe Directional Ontology.** Does moving to H1/H4 rescue the directional edge Program 1 killed at M15? No - zero promotions at either timeframe. **CLOSED.**
- **Program 4 / 4b - Volatility Regime and Transition.** Does contemporaneous volatility-regime *level*, or a forward Markov *transition* forecast, offer a consumable directional edge? Both informative (volatility genuinely has memory) but neither consumable - the transition-forecast variant (4b) turned out to be statistically redundant with plain current-volatility level, adding nothing beyond what was already known. **CLOSED**, both variants.
- **Program 5 - Cross-Sectional Dispersion.** The first program to leave the per-instrument frame entirely and ask whether relative-value/market-neutral positioning across instruments is monetizable. All five interpreters tested REJECT, well-powered. **KILLED.**
- **Program 6 / 6b - Carry / Basis and Carry Harvest.** Is funding-rate carry a usable signal (6), or - separately - is the raw cash-and-carry payoff itself harvestable net of costs (6b)? Both null: all eight carry/basis interpreters REJECT, and all four harvest constructions fail to clear their own turnover cost even before considering price drag. **Both KILLED**, with an explicit stop called after 6b rather than continuing to iterate on cost assumptions.
- **Program 7 - Open Interest.** Named as explicitly out of scope for Program 6 (a distinct payoff), but not yet run as of the evidence this chapter draws on.
- **Program 8 - Weekly Liquidity-Sweep Ontology.** A genuinely new calendar-locked geometry (weekly accumulation/sweep/reversal, distinct from Program 1/2's local-structural frame) tested on FX majors. Zero promotions across all six scopes tested. **CLOSED.**
- **Program 9 - M5-base Multi-TF Expansion Forecasting (Non-Directional Ontology).** See dedicated section below (Grok review pass). **Pre-registered; research authority only.**
- **A cross-asset generalization check (not itself numbered as a Program):** does Program 1's crypto-majors null generalize to FX majors? Yes - the same entry-information null replicated on five FX majors with well-powered, decisively negative toy results.

Program 9 in depth (Grok review pass - closes the coverage gap)

Earlier book drafts only noted Program 9 from commit history. The frozen pre-registration is authoritative: docs/research/preregistration-program-9.md (`../research/preregistration-program-9.md`).

Why it exists. Program 4 (F-040) found expansion forecasting **informative but not consumable** by a spot *directional* consumer. Its reopen clause allows only new data, a new domain, or a new

non-directional ontology - never parameter archaeology. Program 9 exercises two reopen keys:

- 1. **New data:** strict M5 corpus (Binance crypto ~2y; MT5 FX ~270d) that did not exist when F-040 registered ("no M5 on disk").
- 2. **New ontology:** a both-sided stop-entry **OCO straddle** at compression-box edges (long-volatility payoff), which spot long/short cannot express.

Two-stage gate (same Authority Ladder as every other program):

Stage	Question	Authority level
Stage 1	Does the M5^M15^H1^H4 conjunction carry <i>incremental</i> information beyond the M15-base key at the same instant? (Gates A raw IG + B within-K15-cell incremental)	Level 1 - information only
Stage 2	Sole consumer <code>compression_box_straddle</code> through the unchanged M4 QualificationGate	Level 2 - economic

Stage 2 runs **only if** Stage 1 clears. **Gate A pass + Gate B fail** $\Rightarrow$ **M15_REDUNDANT = Stage-1 FAIL** (F-043's "real but already-known" lesson, renamed for this program).

Frozen construction highlights (D1-D7): reject-bar both-edges cancel (no invented touch order); no fill-bar TP credit; unfilled straddles excluded from n; first-compression-bar gating only; chronological 50/50 stability split; sole-consumer rule - no straddle-v2 archaeology inside Program 9.

Closure rule: Program 9 is **CLOSED** once hypotheses 9a (vol expansion), 9b (range expansion), and 9c (compression->expansion transition) each resolve to Stage-1 FAIL (incl. **M15_REDUNDANT**) **or** Stage-1 PASS + Stage-2 sole-consumer verdict - under the fixed thresholds. No within-program iteration.

Status at book time: pre-registration is frozen and present in-repo; hypothesis registry seed tracks H-016 as open/pre-reg. This book does **not** invent Stage-1/Stage-2 numeric results - verify artifacts under `results/research/m5_mtf/` (if present) and `docs/current-findings.md` before claiming a PROMOTE/REJECT. Authority remains **research/docs only** until a sealed finding says otherwise.

Related implementation surfaces (orientation, not a full walk): `src/research/candle_state/` (M5 Stage-1 kernels), `src/research/hypotheses/compression_box_straddle.py` (Stage-2 consumer), `src/research/resample.py` (calendar OHLCV resampler shared with Program 3).

Why this is the right outcome, not a broken pipeline

Every one of these closures follows the same discipline as [Chapter 9's](#) Interpreter Contract: pre-register the test, apply a fixed, honest cost/exit model, require *absolute* expectancy, and accept the answer even when it's negative. A research program that only ever produced PROMOTE verdicts on this codebase's own history would be the actual red flag - it would mean the bar wasn't being held. Programs 1 through 8 closing null, one after another, under the same rigorous standard, is what a functioning falsification discipline looks like from the outside.

Classification

Concept	Status
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Programs 1, 2, 3, 4, 4b, 5, 6, 6b, 8	Research - all CLOSED/KILLED/FROZEN , null or insufficient-power results
Program 9 (M5 multi-TF non-directional straddle)	Research - PRE-REGISTERED ; Stage results not asserted here
Program 7 (Open Interest)	Named / out-of-scope for Program 6; not independently verified as run
FX cross-asset generalization of Program 1's null	Research - confirmed generalizes

Authoritative sources

- `docs/current-findings.md` - the Funding Ledger section, Programs 1 through 8, each with its own F-0xx findings and confidence levels - this chapter's primary source, summarized not reproduced.
- `docs/research/preregistration-program-9.md` - Program 9 frozen pre-registration (primary for P9).
- `docs/research-readiness/program-*-preregistration.md` - the pre-registration for each program (e.g. `program-4-nondirectional-preregistration.md`, `program-8-weekly-crt-sweep-preregistration.md`).
- `CLAUDE.md` §6.5 - the Authority Ladder this whole chapter is an application of.

Unresolved questions

- **Program 9 run artifacts** - whether Stage-1/Stage-2 JSON under `results/research/m5_mtf/` is present and sealed into a finding was not re-validated against a fresh run this pass; the *design* is now chaptered, the *verdict* must still come from artifacts + findings.
- Program 7 (Open Interest) is named but its current status (planned, in progress, or not started) wasn't independently verified this pass.

Previous: [Chapter 18 - A Field Guide to docs/governance/](#) * **Next:** [Chapter 20 - The Research Platform](#)

Related: [Chapter 09 - Interpreters and the Pattern Contract](#) (the same discipline, one interpreter at a time) * [Chapter 16 - Config-First Doctrine](#) (the Authority Ladder)

Memory: none dedicated - see the research-family memory entries indexed in `MEMORY.md` under this session's project files (e.g. cross-sectional, carry-basis, carry-harvest programs).

Chapter 20 - The Research Platform: src/research/

Part VII - Research

Status of this chapter: Written (platform map + artifact index expanded Grok pass)

Why this chapter exists

Chapter 19 covered the *findings* and **numeric results** the research effort produced. This chapter covers the *machinery* that produces them: `src/research/`, the single largest subpackage in the entire codebase (416 files - more than double the next largest), plus its companion pre-registration corpus in `docs/research-readiness/` and the on-disk result trees under `results/research/`.

What problem it solves

Explains why there's so much code here, what shape it takes, **where research configs and outputs live**, and - importantly - flags a real gap between the research platform's ambition and its current Measurement Contract implementation state.

What you need to already know

Chapter 19 - this chapter is that one's supporting infrastructure.
Read the **Research results at a glance** table there before this map.

The idea

Research data map (where things live on disk)

Layer	Path	What you find
Living conclusions	<code>docs/current-findings.md</code>	F-0xx rows with Evidence, Confidence, Status
Pre-registrations	<code>docs/research/preregistration-*.md</code> <code>docs/research-readiness/program-*.preregistration.md</code>	Frozen hypotheses before runs
Research configs	<code>configs/research/research_config*.json</code> <code>configs/research/experiments/</code>	Per program / per-asset research knobs
Measurement profiles	<code>configs/research/measurement_contr</code> <code>mp-*/profiles.json</code>	Schema ready; MC instances not sealed)
Code kernels	<code>src/research/**</code>	Drivers, adapters, controls, consumers
CLI drivers	<code>scripts/research/**</code>	Thin entry points that write artifacts
Run artifacts	<code>results/research/**</code>	JSON/JSONL ledgers (qualification, phase_, regime, m5_mtf, weekly_sweep, h_msip_, ...)
Family registry	<code>docs/governance/research_family_registry</code>	Discovery layer atlas; most cells Contract:UNKNOWN

Rule: a claim without a file under `results/research/` (or an Evidence path in `current-findings.md`) is not a research result - it is a hypothesis.

The shape of `src/research/`

Structured as ~17 subpackages plus loose top-level modules. Map of **what research data each cluster owns**:

Subpackage / module	Research data role	Tied programs
<code>adapters/</code>	Spine / signal sources for research harnesses	P1 spine arm
<code>candle_state/</code>	MTF conjunction, transition targets, M5 kernels	P4 / P9 Stage-1
<code>controls/</code>	Negative controls every economic arm must beat	All M4 programs
<code>hypotheses/</code>	Economic consumers (e.g. <code>compression_box_straddle</code>)	P9 Stage-2
<code>weekly_sweep/</code>	Calendar-locked weekly sweep geometry	P8
<code>clean_labels/</code>	Honest label re-derivation (<code>forward_walk</code> class)	RR / Zone audits
<code>cross_sectional.py</code>	Panel / relative-value kernel	P5 / P6 / P6b
<code>exit_grid.py</code> , <code>costs.py</code>	SL/TP grid + cost model	P1 exit falsification
<code>conditional_entropy_grid.py</code>	Entropy + economics grid	P1 conditional pockets
<code>resample.py</code>	Deterministic OHLCV resampler	P3 HTF, P9 MTF
<code>measurement/</code>	Measurement-contract scaffolding	MC path (OPEN)
<code>ic002_*</code> / <code>ic003_*</code>	Interpreter-candidate shape research	shape / sequence IC tracks
<code>zone_mapping/</code>	Zone assignment research	ZoneGate audits
<code>secondlow_v1/</code> , <code>path/</code> , <code>episodes/</code>	Path / episode / second-low experiments	specialized tracks
<code>synthetic/</code> , <code>envelope_offline/</code> , <code>model_runners/</code> , <code>forensics.py</code>	Synthetic data, offline envelope, model runners, forensics	supporting

Loose shared modules also include: `experiment_spec.py`, `goal_alignment.py`.

The Edge Research Platform (ERP)

`docs/research-readiness/` documents a broader design - the "Edge Research Platform" - covering how research should flow end-to-end: an information-class boundary, a phased plan, a "promise ladder" (explicitly the doctrine-side counterpart of the Authority Ladder from [Chapter 16](#)), and how multiple LLMs collaborate on research work (tying into [Chapter 23](#)). This directory also holds every individual program's pre-registration document (`program-N-*.preregistration.md`) and a series of individual hypothesis pre-registrations (`h-*` files) for smaller, single-hypothesis tests that don't rise to full-Program scale.

An honest gap: the Research Measurement Contract exists on paper, not yet in code

One specific piece is worth flagging precisely because it's easy to miss:
`docs/governance/MEASUREMENT_CONTRACT.md`
defines a frozen (as of 2026-07-10) schema meant to make every research claim's measurement basis - label derivation, cost model, gate mode, clock basis - explicit and comparable across experiments, backed by a 27-seed adversarial mutation test matrix. As of the most recent verification available to this book, **zero MC- * instances have been sealed and none of the 27 probes have been implemented** - every finding in `docs/current-findings.md` currently carries `Contract: UNKNOWN`. This does not invalidate the Program 1-8 findings in [Chapter 19](#) - they were run under their own program-specific rigor (pre-registration, honest costs, absolute-expectancy bar) - but it does mean the *cross-program comparability* layer the Measurement Contract is meant to provide doesn't exist yet. Anyone tempted to compare two programs' results side by side should treat that comparison as informal until this layer is built.

Proposed path to close the gap (Grok review - roadmap, not an implement claim)

The review correctly treats 0 sealed MC- * instances as a **research-infrastructure** gap, not a failure of Programs 1-8. A minimal credible path (still owner-gated; this book does not execute it):

Step	Deliverable	Why it matters
1. Seal one profile-bound MC	One MC- * instance bound to an existing measurement profile (<code>configs/research/measurement_contracts/*.v1.json</code> , MP- * namespace) for a <i>closed</i> program (e.g. Program 8 or F-040's Stage-2)	Proves the schema is fillable from real artifacts, not only from theory
2. Implement E-MT-00 + E-MT-01	First two probes of the 27-seed mutation matrix	Turns "frozen schema" into "adversarially checked schema"
3. Bind registry rows	Update <code>docs/governance/research_family_registry.json</code> objectxlayer cells from <code>Contract:UNKNOWN</code> -> sealed id for the programs just covered	Makes the atlas machine-queryable
4. Stop the bleed	New research programs refuse to register findings without an MC- * id (policy + optional test floor)	Prevents the UNKNOWN set from growing while history is backfilled
5. Backfill selectively	Only re-seal programs when a comparison claim is needed - do not require rewriting all of F-019...F-043 before any new work	Matches L0 sufficiency (\$10 of the contract): hygiene spend is bounded

What this does *not* claim: sealing contracts does not reverse any null result, does not grant fusion/production authority, and does not replace per-program pre-registration. It only makes "these two experiments measured the same thing" a checkable statement.

Reports that check the platform's own hygiene

`docs/research-readiness/` also holds a set of integrity reports that audit the research *infrastructure* itself rather than any one finding: `determinism-report.md` (is the same experiment reproducible), `metric-integrity-report.md`, `config-reachability-report.md`, `test-gap-report.md`, `behavior-census-report.md`. These exist so that a null result can be trusted as a genuine null, rather than an artifact of broken plumbing.

Classification

Concept	Status
src/research/ subpackages	Research infrastructure, actively used (Programs 1-8)
ERP design docs (docs/research-readiness/edge-research-platform-*)	Design doctrine, partially implemented
Research Measurement Contract	Frozen schema, unimplemented - 0/27 probes built, all findings Contract: UNKNOWN
Research-readiness integrity reports	Production tooling (self-audits the research platform)

Authoritative sources

- `src/research/` - the 17 subpackages and shared modules listed above.
- `docs/research-readiness/README.md` and its ~60 sibling files (ERP design series, program and hypothesis pre-registrations, integrity reports).
- `docs/governance/MEASUREMENT_CONTRACT.md` - the frozen, unimplemented contract schema.
- `docs/topics/research-measurement-contract.md` - the always-synced topic doc for this specific gap.

Unresolved questions

- **Sealed mc- * count** - re-verify against `docs/governance/research_family_registry.json` and `MEASUREMENT_CONTRACT.md` before claiming any seal; book snapshot remains `OPEN` / 0 sealed unless those artifacts say otherwise.
- This chapter did not individually verify every one of the 17 `src/research/` subpackages' current wiring status (research-only vs. touching any promotable artifact) - flagged in [A2](#).

Encyclopedia: [E2 - Research Utilities Index](#) (269-file program/driver map)

Previous: [Chapter 19 - The Research Programs](#) * **Next:** [Chapter 21 - The AI Automation Agent](#)

Related: [Chapter 23 - Multi-LLM Coordination](#) (the ERP's multi-LLM research-collaboration design)

Memory: the session memory entry on the research family registry + measurement profiles (title: "Research family registry + measurement profiles") records the Measurement Contract's frozen-since-2026-07-10, zero-implementation status - consult it via the assistant's memory index, not a repo-relative path (it lives outside this repository).

Chapter 09 - Interpreters and the Pattern Contract

Part III - Market Semantics

Status of this chapter: Written

Why this chapter exists

CRT (Chapter 8) is one way of giving candle sequences structural meaning. It is not the only pattern vocabulary a trader might reach for - Point & Figure, Wyckoff, Market Profile, Order Flow are all classical alternatives. This chapter covers how this repository lets such ideas in *without* letting them claim intelligence they haven't earned.

What problem it solves

Prevents "we added a pattern detector" from silently becoming "we trust this pattern detector." Every interpreter has to prove itself the same way, before it can influence anything.

What you need to already know

[Chapter 1](#)'s "measured, never asserted" principle, and
[Chapter 8](#) - interpreters are a parallel idea to CRT, not a replacement.

The idea

The Interpreter Contract (Level 4)

`src/interpreters/contract.py`, `adapter.py`, and `reference.py` define a fixed contract any pattern-recognition module must satisfy before it's allowed to matter. A candidate pattern is wrapped as an `InterpreterHypothesis` and run through the same `forward_walk` evaluation plus `QualificationGate` machinery used elsewhere in the research pipeline - the same yardstick, reused, not a bespoke one invented per interpreter. This is deliberate: it means "does this pattern help" is answered the same way every time, and a pattern can't get a friendlier evaluation just because its author believed in it more.

The one interpreter that's actually been run through it

Point & Figure (P&F, implementation id `PNF-v1`, double-top/double-bottom patterns) is the first interpreter to complete this process on real data. The result was a clean **REJECT**: shadow-measured on crypto majors, it carried no standalone edge, and scored worse than random controls (a negative delta-expectancy of about -0.039). This is reported here not as a failure of the interpreter framework but as evidence the framework *works as intended* - it correctly declined to grant authority to an idea that didn't earn it, exactly the discipline [Chapter 1](#) described. Per the repository's own research discipline, this closes P&F under the current ontology; reopening it needs a genuinely new ontology, not another sweep of the same one.

What this means for other interpreters

Nothing else currently in `src/interpreters/` has cleared the contract - until one does, the honest status of the whole subsystem is: infrastructure exists, one candidate has been tested and rejected, the rest are unevaluated. None of them influence the live decision spine.

Classification

Concept	Status
Interpreter Contract (<code>contract.py/adapter.py/reference.py</code>)	Production infrastructure
P&F / PNF - v1	Research - tested, REJECTED (F-028)
Wyckoff, Market Profile, Order Flow interpreters	Unknown / not yet run through the contract

Authoritative sources

- `src/interpreters/contract.py`, `adapter.py`, `reference.py` - the contract itself.
- `docs/topics/interpreter-contract.md` - the always-synced topic doc, entry point, and test references.
- `docs/current-findings.md` F-028 - the P&F result, with its exact scope and confidence.
- `docs/architecture/goal.md` - the "measured against the Goal Layer, never asserted" framing this chapter draws its principle from.

Unresolved questions

- Whether Wyckoff/Market Profile/Order Flow interpreters have any code beyond scaffolding, and if so how far along the contract pipeline they are, wasn't verified in this pass - flagged in [A2](#).

Previous: [Chapter 08 - The CRT State Machine](#) * **Next:** [Chapter 10 - The Four Scoring Engines](#)

Related: [Chapter 19 - The Research Programs](#) (the same falsification discipline, at system scale)

Memory: `docs/memory/engine-memory.md`.

Appendix A2 - Unresolved Questions (Rollup)

Status of this chapter: Written (updated Grok review pass, 2026-08-07)

Why this chapter exists

Per this book's own charter: "if understanding is impossible, do not invent." Every chapter that hit a genuine gap in evidence recorded it locally rather than guessing. This appendix collects those in one place - and, after the Grok review pass, also records what was **closed in the book** without changing production code.

What you need to already know

This is a reference appendix, not a narrative chapter - each item links back to the chapter that owns the full context.

Resolved in the book (Grok review pass - no code changes)

Item	Resolution	Chapter
INOUT <code>src/inout/</code> vs <code>archive/inout_legacy/</code>	Name collision. Current <code>src/inout/</code> = data fetchers only; archived tree = parallel execution rail retired 2026-05-02	Ch.15
Program 9 missing from research chapter	Chaptered from pre-registration (<code>docs/research/preregistration-program-9.md</code>); run verdicts still artifact-bound	Ch.19
ATR / F-057 "flag only" passivity	Proposed remediation options + trade-offs added; still OPEN pending owner	Ch.13
Measurement Contract gap	5-step seal path proposed; still 0 sealed instances in-repo unless registry says otherwise	Ch.20
MSIP "no treatment"	Orientation brief in governance field guide	Ch.18
Density / meta overwhelm	Quick Start chapter added	Ch.00
22 un-chaptered packages	Orientation map added (not full chapters)	Ch.04

Still open - book-scope gaps

- **Repository Encyclopedia** - charter in [Chapter 24](#);
E0-E6 + E1b + JSONL twin DONE (`docs/book/encyclopedia/`, including `encyclopedia_rows.jsonl`). Optional later: relationship graph edges in JSONL; agent query UI.
- **Full-depth Part VI-VIII** - chapters 16-23 remain orientation-level relative to their source corpora (~250 governance files, ~475 research files). Deepening is expected future book work.
- **Optional MSIP dedicated chapter** - brief exists; full gate-by-gate narrative only if MSIP re-enters active execution.
- **Program 9 numeric verdict** - design is chaptered; Stage-1/Stage-2 results must still be read from artifacts / findings, not assumed.

- **Program 7 (Open Interest)** - named as out-of-scope for Program 6; status not independently verified as run.
- **Deeper Part VIII implementation walk** - agent/control-plane/multi-LLM chapters improved slightly but are not full replacements for docs/reference/agent-reference.md.

Still open - real repository issues (code/doc; not fixed by this book pass)

User instruction for this pass: **no code changes**. These remain tracked, not remediated.

- docs/reference/schemas.md §4 still documents the 38-dim v3.0 feature tuple; live code is 39-dim v4.0. Textbook DOC_DRIFT. See [Chapter 7](#). Recommended separate Doc Sync turn - not part of this PDF/book narrative pass.
- docs/reference/testing.md file-count header ("116 files across 15 directories," dated 2026-06-12) is stale vs ~1,200+ files under tests/. See [Appendix A1](#).
- ATR unit mismatch in compute_crt_levels - real geometry defect; options A/B/C in [Chapter 13](#); awaiting owner decision.
- F-057 crtconfig programmatic split-brain - OPEN; options 1/2/3 in Chapter 13; "backtest Phase 2."
- Research Measurement Contract - schema frozen, 0 sealed MC-* (verify registry); path in [Chapter 20](#).

Verification gaps (assumed from docs; not re-proven this pass)

- Whether Wyckoff / Market Profile / Order Flow interpreters have code beyond scaffolding ([Chapter 9](#)).
- Whether every tool in docs/reference/agent-reference.md is end-to-end reachable ([Chapter 21](#)).
- Full body of docs/reference/architecture.md beyond header/orientation ([Chapter 4](#)).

Out of scope for this pass (explicit)

- Fixing schemas.md / testing.md primary docs (Doc Sync turn).
- Implementing ATR or F-057 remediations (code + parity + promotion).
- Sealing MC-* instances or running Program 9 Stage drivers.
- Adding CLAUDE.md companion-row (already present from an earlier session).

Previous: [Appendix A1 - Testing the System](#) * **Next:** - (last chapter; return to [the Table of Contents](#))

Related: every chapter linked above.

Memory: none dedicated - this appendix is the book's own open-loop tracker.